

B6 Condensed-Matter Physics — Model Solutions, 2018

Question 1 — Crystal structure of tungsten

Problem. Identify lattice types of α - and β -tungsten; from the lowest-angle powder XRD ring of α -W (0.1542 nm X-rays, $2\theta = 40.26^\circ$) deduce a and the density; for β -W find the first two non-vanishing rings, their intensity ratio, and the lowest-angle additional weak ring expected.

(a) Definition and lattice type of α -W

A *lattice* is the infinite set of points

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3, \quad n_i \in \mathbb{Z},$$

generated by three primitive vectors $\mathbf{a}_i$; every lattice point has identical environment.

For α -W, the two atoms at $[0, 0, 0]$ and $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ are chemically identical and have identical local environments, so each is a lattice point. The conventional cubic cell with the centring vector $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ is

body-centred cubic (BCC).

(b) Lattice constant and density

Selection rule for BCC: reflections allowed when $h + k + l$ is even.

Identify the lowest-sin θ ring:

$$\text{lowest-angle ring} = (110), \quad h^2 + k^2 + l^2 = 2.$$

Bragg's law:

$$\lambda = 2d_{110} \sin \theta.$$

Solve for d_{110} :

$$d_{110} = \frac{\lambda}{2 \sin \theta}.$$

Halve the scattering angle $2\theta = 40.26^\circ$:

$$\theta = 20.13^\circ.$$

Evaluate $\sin \theta$:

$$\sin(20.13^\circ) = 0.3441.$$

Substitute the numerical values:

$$d_{110} = \frac{0.1542 \text{ nm}}{2 \times 0.3441}.$$

Evaluate:

$$d_{110} = 0.2240 \text{ nm}.$$

Cubic spacing $d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$ gives:

$$a = d_{110}\sqrt{2}.$$

Substitute:

$$a = (0.2240 \text{ nm})\sqrt{2}.$$

Evaluate:

$$a = 0.3168 \text{ nm}.$$

BCC conventional cell contains 2 atoms. Mass density:

$$\rho = \frac{2M}{N_A a^3}.$$

Substitute the numerical values:

$$\rho = \frac{2 \times 0.1838 \text{ kg mol}^{-1}}{(6.022 \times 10^{23} \text{ mol}^{-1})(3.168 \times 10^{-10} \text{ m})^3}.$$

Evaluate a^3 :

$$a^3 = 3.179 \times 10^{-29} \text{ m}^3.$$

Combine numerator and denominator:

$$\rho = \frac{0.3676 \text{ kg mol}^{-1}}{1.914 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}}.$$

Evaluate:

$$\rho \approx 1.92 \times 10^4 \text{ kg m}^{-3} = 19.2 \text{ g cm}^{-3}.$$

(c) Lattice type of β -W

The two corner/centre atoms $\{[0, 0, 0], [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]\}$ have a different local environment from the six face-pair atoms $[\frac{1}{4}, \frac{1}{2}, 0]$ etc., so the eight sites are *not* all equivalent: they cannot all be lattice points.

Test BCC centring $\mathbf{T} = [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$:

$$[\frac{1}{4}, \frac{1}{2}, 0] + [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}] = [\frac{3}{4}, 1, \frac{1}{2}] \equiv [\frac{3}{4}, 0, \frac{1}{2}],$$

which is not in the basis: BCC ruled out. FCC and primitive face centring similarly fail. Hence

simple (primitive) cubic, with an 8-atom basis (A15 structure).

(d) First two rings of β -W and their intensity ratio

All eight atoms have the same form factor f . Structure factor:

$$S_{hkl} = f \sum_{j=1}^8 e^{2\pi i(hx_j + ky_j + lz_j)}.$$

Corner pair $[0, 0, 0], [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$:

$$S_c/f = 1 + e^{i\pi(h+k+l)}.$$

Use $e^{i\pi n} = (-1)^n$:

$$S_c/f = 1 + (-1)^{h+k+l}.$$

Face pair on the $z = 0$ face, $[\frac{1}{4}, \frac{1}{2}, 0] + [\frac{3}{4}, \frac{1}{2}, 0]$:

$$S_{f,z}/f = e^{2\pi i(h/4+k/2)} + e^{2\pi i(3h/4+k/2)}.$$

Factor out $e^{i\pi k}$:

$$S_{f,z}/f = e^{i\pi k} (e^{i\pi h/2} + e^{i3\pi h/2}).$$

Factor $e^{i\pi h}$ from the bracket:

$$e^{i\pi h/2} + e^{i3\pi h/2} = e^{i\pi h} (e^{-i\pi h/2} + e^{i\pi h/2}).$$

Apply $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$:

$$e^{i\pi h/2} + e^{i3\pi h/2} = 2e^{i\pi h} \cos(\pi h/2).$$

Substitute back:

$$S_{f,z}/f = 2(-1)^{h+k} \cos(\pi h/2).$$

Repeat the same factorisation for the pairs on the $y = 0$ face ($[\frac{1}{2}, 0, \frac{1}{4}], [\frac{1}{2}, 0, \frac{3}{4}]$) and on the $x = 0$ face ($[0, \frac{1}{4}, \frac{1}{2}], [0, \frac{3}{4}, \frac{1}{2}]$). Sum the three face-pair contributions:

$$S_f/f = 2(-1)^{h+k} \cos \frac{\pi h}{2} + 2(-1)^{h+l} \cos \frac{\pi l}{2} + 2(-1)^{k+l} \cos \frac{\pi k}{2}.$$

Combine corner and face contributions:

$$S/f = [1 + (-1)^{h+k+l}] + 2[(-1)^{h+k} \cos \frac{\pi h}{2} + (-1)^{h+l} \cos \frac{\pi l}{2} + (-1)^{k+l} \cos \frac{\pi k}{2}].$$

Evaluate for low (hkl) :

(hkl)	$h^2 + k^2 + l^2$	S/f	m
(100)	1	$0 + 2[0 - 1 + 1] = 0$	—
(110)	2	$2 + 2[0 - 1 + 0] = 0$	—
(111)	3	$0 + 2[0 + 0 + 0] = 0$	—
(200)	4	$2 + 2[-1 + 1 + 1] = 4$	6
(210)	5	$0 + 2[1 + 1 + 0] = 4$	24
(211)	6	$2 + 2[1 + 0 + 0] = 4$	24
(220)	8	$2 + 2[-1 + 1 - 1] = 0$	—

First two non-vanishing rings:

$(200) \text{ and } (210).$

Both have $|S|^2 = 16f^2$, so the ratio is set by multiplicity (Lorentz-polarisation and Debye-Waller corrections are subleading):

$$\frac{I_{210}}{I_{200}} \approx \frac{m_{210}}{m_{200}} = \frac{24}{6}.$$

$\frac{I_{210}}{I_{200}} \approx 4.$

(e) Additional weak ring at lower angle

The cancellations producing $|S| = 0$ assumed all 8 sites scatter identically. The two Wyckoff sites are crystallographically distinct (corner sublattice vs. face-pair sublattice): they vibrate differently, giving distinct anisotropic Debye–Waller factors and slightly different effective form factors, $f_1 \neq f_2$.

Write $S = f_1 S_c + f_2 S_f$ with $S_c = 1 + (-1)^{h+k+l}$ and S_f the face-pair sum. Tabulating (S_c, S_f) for the previously forbidden rings:

$$(100) : (0, 0), \quad (110) : (2, -2), \quad (111) : (0, 0), \quad (220) : (2, -2).$$

Rings with $S_c = S_f = 0$ remain extinct; rings with $S_c \neq 0$ and $S_f \neq 0$ acquire weight $\propto (f_1 - f_2)^2$.

Lowest such ring (smallest $h^2 + k^2 + l^2$):

$(110),$

seen weakly because $|f_1 - f_2| \ll f$.

Question 2 — Tight binding

Problem. Define μ and E_F . For a 1D tight-binding chain with hopping $-t$, derive the dispersion, sketch it and the DOS. For monovalent atoms find E_F and show $\mu(T) = E_F$. Repeat the Fermi-surface and $\mu(T)$ analysis on a 2D square lattice, then on a squeezed rectangular lattice.

(a) Definitions

Chemical potential: $\mu = (\partial U / \partial N)_{S,V}$, the energy cost of adding one particle at fixed entropy and volume; equivalently, the energy at which the Fermi–Dirac function equals $\frac{1}{2}$.

Fermi energy: $E_F = \mu(T = 0)$, the energy of the highest occupied single-particle state in the ground state of a non-interacting fermion system.

(b) 1D dispersion, BZ diagram, DOS

Expand a single-particle eigenstate $|\psi\rangle = \sum_n \psi_n |n\rangle$. Apply H :

$$H |\psi\rangle = \sum_n \psi_n (\epsilon_0 |n\rangle - t |n+1\rangle - t |n-1\rangle).$$

Project onto $\langle m|$ (use $\langle m|n\rangle = \delta_{mn}$):

$$E \psi_m = \epsilon_0 \psi_m - t \psi_{m-1} - t \psi_{m+1}.$$

Substitute Bloch ansatz $\psi_n = e^{ikna}$:

$$E e^{ikma} = \epsilon_0 e^{ikma} - t e^{ik(m-1)a} - t e^{ik(m+1)a}.$$

Divide both sides by e^{ikma} :

$$E = \epsilon_0 - t e^{-ika} - t e^{ika}.$$

Combine exponentials with $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$:

$E(k) = \epsilon_0 - 2t \cos(ka), \quad k \in (-\pi/a, \pi/a].$

DOS per spin in 1D: states in length L between k and $k + dk$ number $(L/2\pi)dk$. With two k branches ($\pm k$) at each E :

$$g(E) dE = 2 \cdot \frac{L}{2\pi} |dk| = \frac{L}{\pi} \frac{dE}{|dE/dk|}.$$

Compute $|dE/dk|$ from the dispersion:

$$\left| \frac{dE}{dk} \right| = 2ta |\sin(ka)|.$$

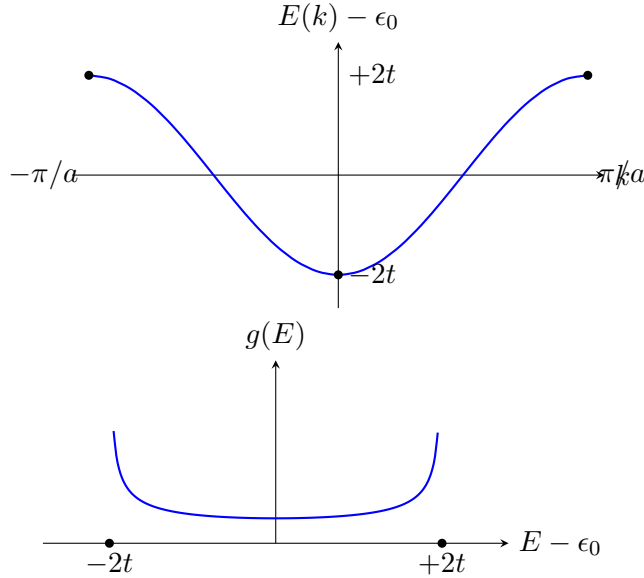
Use $\sin^2(ka) = 1 - \cos^2(ka)$ with $\cos(ka) = (\epsilon_0 - E)/(2t)$:

$$|\sin(ka)| = \frac{1}{2t} \sqrt{(2t)^2 - (E - \epsilon_0)^2}.$$

Substitute and invert:

$$g(E) = \frac{L}{\pi a} \frac{1}{\sqrt{(2t)^2 - (E - \epsilon_0)^2}}.$$

Van Hove singularities at $E = \epsilon_0 \pm 2t$.



(c) Half-filling: E_F and $\mu(T)$

Bandwidth $4t$, N atoms, $2N$ spin states, N electrons (monovalent): half-filled. Symmetry $E(k + \pi/a) - \epsilon_0 = -[E(k) - \epsilon_0]$ gives $g(\epsilon_0 + \delta) = g(\epsilon_0 - \delta)$.

At $T = 0$, fill the lower half of the band:

$$\boxed{E_F = \epsilon_0.}$$

Particle-number constraint per spin at temperature T :

$$N_e^{(\sigma)} = \int g(E) n_F(E, \mu, T) dE.$$

Half-filling sets the per-spin filling to half the band:

$$N_e^{(\sigma)} = \frac{1}{2} \int g(E) dE.$$

Apply the hint $n_F(E, \mu) = 1 - n_F(2\mu - E, \mu)$:

$$\int g(E) n_F(E, \mu) dE = \int g(E) dE - \int g(E) n_F(2\mu - E, \mu) dE.$$

Change variables $E' = 2\mu - E$ in the second integral:

$$\int g(E) n_F(2\mu - E) dE = \int g(2\mu - E') n_F(E') dE'.$$

Substitute back (rename $E' \rightarrow E$):

$$\frac{1}{2} \int g(E) dE = \int g(E) dE - \int g(2\mu - E) n_F(E, \mu) dE.$$

The DOS satisfies $g(2\epsilon_0 - E) = g(E)$ by the $E \rightarrow 2\epsilon_0 - E$ symmetry of the cosine band. Demand consistency by setting $\mu = \epsilon_0$:

$$\int g(2\epsilon_0 - E) n_F(E, \epsilon_0) dE = \int g(E) n_F(E, \epsilon_0) dE = N_e = \frac{1}{2} \int g dE.$$

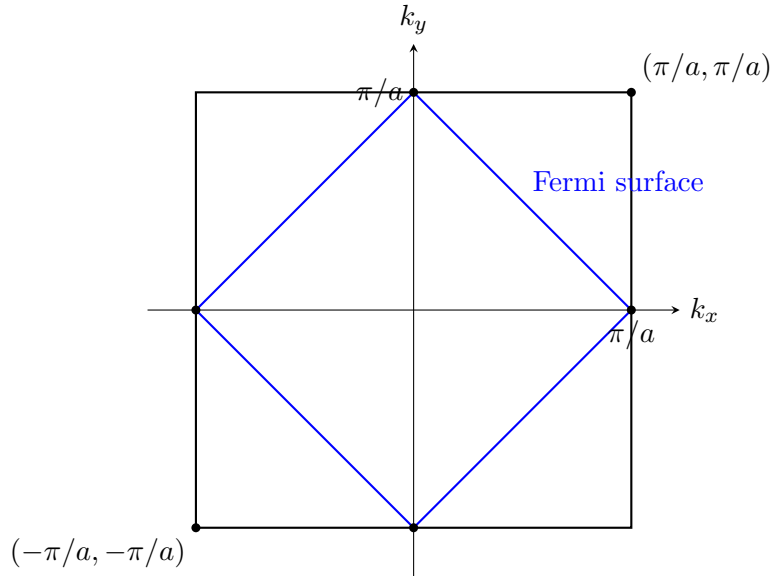
The constraint is then identically satisfied, fixing:

$$\boxed{\mu(T) = \epsilon_0 = E_F \quad \text{for all } T.}$$

Particle-hole symmetry pins μ to the band centre.

(d) 2D square lattice, monovalent

Brillouin zone: square $-\pi/a < k_x, k_y \leq \pi/a$. Half-filling sets $\cos(k_x a) + \cos(k_y a) = 0$: a square rotated by 45° , vertices $(\pm\pi/a, 0), (0, \pm\pi/a)$.



Nesting $E(\mathbf{k} + \mathbf{Q}) + E(\mathbf{k}) = 2\epsilon_0$ with $\mathbf{Q} = (\pi/a, \pi/a)$ gives the same particle-hole symmetry as in (c):

$$\boxed{\mu(T) = \epsilon_0 \quad \text{for all } T.}$$

(e) **Squeezed rectangular lattice**, $a_x < a_y$

Brillouin zone is rectangular: $-\pi/a_i < k_i \leq \pi/a_i$, wider along k_x since $\pi/a_x > \pi/a_y$. Anisotropic dispersion:

$$E(k_x, k_y) = \epsilon_0 - 2t_x \cos(k_x a_x) - 2t_y \cos(k_y a_y).$$

Shorter bond gives stronger overlap so $t_x > t_y$. Half-filling sits at $E = \epsilon_0$:

$$t_x \cos(k_x a_x) + t_y \cos(k_y a_y) = 0.$$

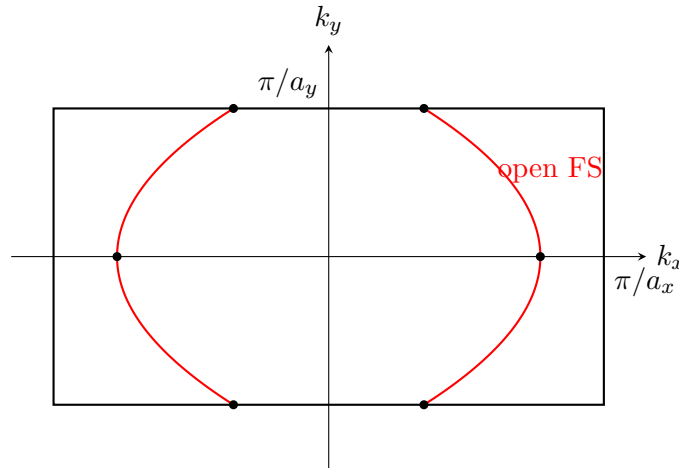
On the $k_y = 0$ line, solve $\cos(k_x a_x) = -t_y/t_x$. Since $|t_y/t_x| < 1$:

$$k_x = \frac{1}{a_x} \arccos\left(-\frac{t_y}{t_x}\right) < \frac{\pi}{a_x}.$$

On the $k_x = 0$ line, $\cos(k_y a_y) = -t_x/t_y$. Since $|t_x/t_y| > 1$ there is no real solution: the Fermi surface does not cross the k_y axis. Instead it reaches the BZ edge $k_y = \pi/a_y$ at:

$$k_x = \frac{1}{a_x} \arccos\left(\frac{t_y}{t_x}\right).$$

Consequently the Fermi surface is *open*: it runs across the BZ in the k_x sense, exiting at $k_y = \pm\pi/a_y$ and reconnecting by zone-boundary periodicity.



Nesting at $\mathbf{Q} = (\pi/a_x, \pi/a_y)$ still gives $E(\mathbf{k} + \mathbf{Q}) + E(\mathbf{k}) = 2\epsilon_0$, so the particle-hole argument of (c) survives:

$$\mu(T) = \epsilon_0 \quad \text{for all } T.$$

Question 3 — Phonons in lead

Problem. Define phonon and Brillouin zone. From the Morse potential ($\xi = 0.084 \text{ nm}$, $A = 1600 \text{ eV}$, $B = 40 \text{ eV}$) compute the equilibrium bond length a , binding energy and harmonic spring constant κ . Compute the 1D sound velocity (Pb, $M = 207 \text{ g mol}^{-1}$). Estimate the thermal-expansion coefficient using cubic anharmonicity, and identify the validity ranges in T .

(a) Definitions

Phonon: a quantised normal mode of lattice vibration carrying energy $\hbar\omega$ and crystal momentum $\hbar\mathbf{k}$, obeying Bose–Einstein statistics.

Brillouin zone: the Wigner–Seitz primitive cell of the reciprocal lattice; equivalently, the set of $\mathbf{k}$ closer to the origin than to any other reciprocal lattice point.

(b) Morse: $a, V_{\min}, \kappa$

Set $u \equiv e^{-r/\xi}$:

$$V = Au^2 - Bu.$$

Apply chain rule with $du/dr = -u/\xi$:

$$V'(r) = (2Au - B) \frac{du}{dr}.$$

Substitute du/dr :

$$V'(r) = -\frac{u}{\xi}(2Au - B).$$

Equilibrium condition $V'(a) = 0$ with $u_a \neq 0$:

$$2Au_a - B = 0.$$

Solve for u_a :

$$u_a = \frac{B}{2A}.$$

Invert $u_a = e^{-a/\xi}$:

$$a = \xi \ln \frac{2A}{B}.$$

Substitute numbers, $2A/B = 80$:

$$a = (0.084 \text{ nm}) \ln 80.$$

Evaluate $\ln 80 = 4.382$:

$a = 0.368 \text{ nm}.$

Binding energy: insert $u_a = B/(2A)$ into V :

$$V(a) = A \left(\frac{B}{2A} \right)^2 - B \cdot \frac{B}{2A}.$$

Simplify each term:

$$V(a) = \frac{B^2}{4A} - \frac{B^2}{2A}.$$

Combine:

$$V(a) = -\frac{B^2}{4A}.$$

Substitute numbers:

$V(a) = -\frac{(40)^2}{4 \cdot 1600} \text{ eV} = -0.250 \text{ eV}.$

Second derivative. Differentiate $V'(r) = -(u/\xi)(2Au - B)$:

$$V''(r) = -\frac{1}{\xi} \left[\frac{du}{dr}(2Au - B) + 2Au \frac{du}{dr} \right].$$

Substitute $du/dr = -u/\xi$:

$$V''(r) = \frac{u}{\xi^2}(4Au - B).$$

Evaluate at $u = u_a = B/(2A)$:

$$\kappa \equiv V''(a) = \frac{B}{2A\xi^2} \left(4A \cdot \frac{B}{2A} - B \right).$$

Simplify the bracket:

$$\kappa = \frac{B}{2A\xi^2} B = \frac{B^2}{2A\xi^2}.$$

Substitute numbers (in eV per nm²):

$$\kappa = \frac{40^2}{2 \cdot 1600 \cdot (0.084)^2} \text{ eV nm}^{-2}.$$

Evaluate:

$$\kappa = 70.9 \text{ eV nm}^{-2}.$$

Convert to SI via $1 \text{ eV nm}^{-2} = 0.1602 \text{ N m}^{-1}$:

$\kappa \approx 11.4 \text{ N m}^{-1}.$

(c) Sound velocity

Monatomic chain dispersion:

$$\omega(k) = 2\sqrt{\kappa/m} |\sin(ka/2)|.$$

Long-wavelength group velocity:

$$v_s = \left. \frac{d\omega}{dk} \right|_{k \rightarrow 0} = 2\sqrt{\kappa/m} \cdot \frac{a}{2}.$$

Simplify:

$$v_s = a\sqrt{\kappa/m}.$$

Atomic mass:

$$m = \frac{M}{N_A} = \frac{0.207 \text{ kg mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}}.$$

Evaluate:

$$m = 3.44 \times 10^{-25} \text{ kg}.$$

Compute κ/m :

$$\frac{\kappa}{m} = \frac{11.4}{3.44 \times 10^{-25}} \text{ s}^{-2} = 3.31 \times 10^{25} \text{ s}^{-2}.$$

Take the square root:

$$\sqrt{\kappa/m} = 5.76 \times 10^{12} \text{ s}^{-1}.$$

Multiply by a :

$$v_s = (0.368 \times 10^{-9} \text{ m}) \times (5.76 \times 10^{12} \text{ s}^{-1}).$$

Evaluate:

$v_s \approx 2.12 \times 10^3 \text{ m s}^{-1}.$

Compare to measured Pb longitudinal speed $\sim 2000 \text{ m s}^{-1}$: agreement.

(d) Thermal expansion from cubic anharmonicity

Expand V to cubic order around $r = a$, with $x \equiv r - a$:

$$V(r) \approx V(a) + \frac{1}{2}\kappa x^2 + \frac{1}{6}\kappa_3 x^3, \quad \kappa_3 \equiv V'''(a).$$

Differentiate V'' once more:

$$V'''(r) = -\frac{1}{\xi^3}(8Au^2 - Bu).$$

Insert $u_a = B/(2A)$:

$$\kappa_3 = -\frac{1}{\xi^3} \left(\frac{2B^2}{A} - \frac{B^2}{2A} \right) = -\frac{3B^2}{2A\xi^3} < 0.$$

Treat cubic term as perturbation; harmonic averages:

$$\langle x \rangle_0 = 0, \quad \langle x^2 \rangle_0 = \frac{k_B T}{\kappa}, \quad \langle x^3 \rangle_0 = 0, \quad \langle x^4 \rangle_0 = 3 \left(\frac{k_B T}{\kappa} \right)^2.$$

First-order shift from the cubic term:

$$\langle x \rangle \approx -\frac{\kappa_3}{6k_B T} \langle x^4 \rangle_0.$$

Substitute $\langle x^4 \rangle_0$:

$$\langle x \rangle = -\frac{\kappa_3}{6k_B T} \cdot 3 \left(\frac{k_B T}{\kappa} \right)^2.$$

Simplify:

$$\langle x \rangle = -\frac{\kappa_3 k_B T}{2\kappa^2}.$$

Differentiate w.r.t. T and divide by a :

$$\boxed{\alpha = \frac{1}{a} \frac{d\langle x \rangle}{dT} \sim \frac{|\kappa_3| k_B}{\kappa^2 a}}.$$

Numerical sanity check: $|\kappa_3| = 3B^2/(2A\xi^3)$. Compute in eV/nm³:

$$|\kappa_3| = \frac{3 \cdot 40^2}{2 \cdot 1600 \cdot (0.084)^3} \text{ eV nm}^{-3} = 2.53 \times 10^3 \text{ eV nm}^{-3}.$$

Convert to SI (multiply by $1.602 \times 10^{-19}/10^{-27} = 1.602 \times 10^8$):

$$|\kappa_3| \approx 4.1 \times 10^{11} \text{ J m}^{-3} = 4.1 \times 10^{11} \text{ N m}^{-2}.$$

Substitute with $\kappa = 11.4 \text{ N m}^{-1}$, $a = 0.368 \text{ nm}$, $k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$:

$$\alpha \sim \frac{(4.1 \times 10^{11} \text{ N m}^{-2})(1.381 \times 10^{-23} \text{ J K}^{-1})}{(11.4 \text{ N m}^{-1})^2 (3.68 \times 10^{-10} \text{ m})}.$$

Evaluate:

$$\alpha \sim 1.2 \times 10^{-4} \text{ K}^{-1}.$$

Order-of-magnitude correct (measured $\alpha_{\text{Pb}} \approx 2.9 \times 10^{-5} \text{ K}^{-1}$); positive sign follows because $\kappa_3 < 0$ widens the well outwards.

(e) Validity ranges

(i) Cubic-only expansion. Require harmonic $\gg$ cubic energy at thermal amplitude $|x| \sim \sqrt{k_B T / \kappa}$:

$$\frac{1}{2} \kappa x^2 \gg \frac{1}{6} |\kappa_3| x^3.$$

Cancel one x^2 and reorganise:

$$\kappa \gg \frac{1}{3} |\kappa_3| x.$$

Substitute $|x| \sim \sqrt{k_B T / \kappa}$:

$$\kappa \gg \frac{|\kappa_3|}{3} \sqrt{k_B T / \kappa}.$$

Square and rearrange:

$$k_B T \ll \frac{9\kappa^3}{\kappa_3^2}.$$

Substitute numerical values:

$$\frac{9\kappa^3}{\kappa_3^2} = \frac{9 \cdot (11.4)^3}{(4.1 \times 10^{11})^2} \text{ J} \sim 8 \times 10^{-20} \text{ J}.$$

Divide by k_B :

$$T \ll 8 \times 10^{-20} \text{ J} / (1.38 \times 10^{-23} \text{ J K}^{-1}) \sim 6 \times 10^3 \text{ K}.$$

The physical bound is set well below this by the lead melting point $T_m \approx 600 \text{ K}$:

$$\boxed{T \lesssim T_m \approx 600 \text{ K}.$$

(ii) Quantum mechanics important. Define crossover $k_B T_Q \sim \hbar \omega$ with $\omega = \sqrt{\kappa/m} = 5.76 \times 10^{12} \text{ s}^{-1}$:

$$T_Q \sim \frac{\hbar \omega}{k_B}.$$

Substitute numerical values:

$$T_Q \sim \frac{(1.055 \times 10^{-34} \text{ J s})(5.76 \times 10^{12} \text{ s}^{-1})}{1.381 \times 10^{-23} \text{ J K}^{-1}}.$$

Evaluate:

$$\boxed{T_Q \approx 44 \text{ K}.$$

For $T \lesssim T_Q$ zero-point motion dominates ($\langle x^2 \rangle \rightarrow \hbar/(2m\omega)$), the harmonic equipartition $\langle x^2 \rangle = k_B T / \kappa$ fails, the heat capacity drops below k_B/mode (Debye/Einstein freeze-out), and $\alpha \propto T^3$ at low T (Grüneisen).

The classical cubic-only treatment is therefore valid for $T_Q \lesssim T \lesssim T_m$, i.e. roughly $44 \text{ K} \lesssim T \lesssim 600 \text{ K}$.

Question 4 — Magnetism of GdN

Problem. Define paramagnet, diamagnet, ferromagnet (with examples). Find S of Gd^{3+} ($4f^7$) by Hund. Treat Gd^{3+} as Ising along $\hat{z}$ in FCC GdN ($a = 0.50 \text{ nm}$): compute Curie constant C for $\chi = C/T$, modify χ for $T > T_c = 70 \text{ K}$, give $M(T = 0)$, and sketch $M(T)$ for $0 \leq T < 2T_c$ at $H_z = 0$ and $H_z > 0$.

(a) Definitions

Paramagnet: $\chi > 0$, no spontaneous order; high- T Curie law $\chi \propto 1/T$. Example: MnSO_4 (or any rare-earth salt above its ordering T).

Diamagnet: $\chi < 0$, repelled from field, Larmor response of closed shells. Example: Cu, Bi, water.

Ferromagnet: spontaneous magnetisation $M \neq 0$ below Curie T_c . Example: Fe, Ni, Co.

(b) Spin of Gd^{3+}

Configuration $4f^7$: half-filled shell. Hund's first rule maximises S , all 7 spins parallel:

$$S = 7/2.$$

(Hund 2: $L = 0$; Hund 3: $J = S = 7/2$, $g_J = 2$.)

(c) Curie constant

Ising along $\hat{z}$ with $S = 7/2$ and $g = 2$: each ion is a two-level system. Moment per ion:

$$m = g S \mu_B = 2 \cdot \frac{7}{2} \mu_B.$$

Simplify:

$$m = 7\mu_B.$$

Zeeman energies in field $B = \mu_0 H_z$ with $m_z = \pm m$:

$$E_{\pm} = \mp mB.$$

Two-level partition function:

$$Z = e^{\beta mB} + e^{-\beta mB} = 2 \cosh(\beta mB).$$

Thermal-average moment:

$$\langle m_z \rangle = m \tanh(\beta mB).$$

Linearise for $\beta mB \ll 1$ using $\tanh x \approx x$:

$$\langle m_z \rangle \approx \beta m^2 B = \frac{m^2 \mu_0 H_z}{k_B T}.$$

Multiply by number density:

$$M_z = n \langle m_z \rangle = \frac{n m^2 \mu_0}{k_B T} H_z.$$

Read off $\chi = M_z/H_z$ and C :

$$\chi = \frac{n m^2 \mu_0}{k_B T} = \frac{C}{T}, \quad C = \frac{n m^2 \mu_0}{k_B}.$$

Number density of Gd^{3+} : FCC conventional cell contains 4 formula units:

$$n = \frac{4}{a^3} = \frac{4}{(0.50 \times 10^{-9} \text{ m})^3}.$$

Evaluate:

$$n = 3.20 \times 10^{28} \text{ m}^{-3}.$$

Compute $(7\mu_B)^2$ with $\mu_B = 9.274 \times 10^{-24} \text{ J T}^{-1}$:

$$m^2 = (6.492 \times 10^{-23} \text{ J T}^{-1})^2 = 4.215 \times 10^{-45} \text{ J}^2 \text{ T}^{-2}.$$

Substitute numerical values into C (with $\mu_0 = 1.257 \times 10^{-6} \text{ T m A}^{-1}$):

$$C = \frac{(3.20 \times 10^{28} \text{ m}^{-3})(4.215 \times 10^{-45} \text{ J}^2 \text{ T}^{-2})(1.257 \times 10^{-6} \text{ T m A}^{-1})}{1.381 \times 10^{-23} \text{ J K}^{-1}}.$$

Evaluate the numerator:

$$nm^2\mu_0 = 1.696 \times 10^{-22} \text{ J}.$$

Divide by k_B :

$$C \approx 12.3 \text{ K}.$$

(d) Curie–Weiss modification and $M(0)$

Mean-field treatment of exchange: replace T by $T - T_c$:

$$\chi(T) = \frac{C}{T - T_c} \quad \text{for } T > T_c.$$

Denominator vanishes at T_c , signalling spontaneous order.

At $T = 0$ all Ising spins align with the field:

$$M(0) = nm = 7n\mu_B.$$

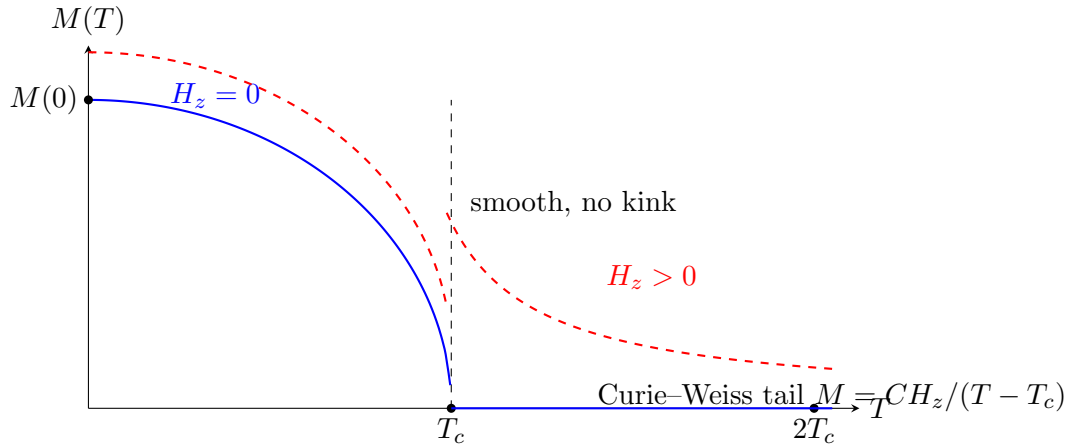
Substitute numbers:

$$M(0) = (3.20 \times 10^{28} \text{ m}^{-3}) \times 7 \times (9.274 \times 10^{-24} \text{ J T}^{-1}).$$

Evaluate:

$$M(0) \approx 2.08 \times 10^6 \text{ A m}^{-1}.$$

(e) $M(T)$ sketch



Differences:

- $H_z = 0$: spontaneous M only for $T < T_c$; $M \propto (T_c - T)^{1/2}$ near T_c in mean-field; identically zero for $T \geq T_c$.
- $H_z > 0$: field selects the $+\hat{z}$ branch, giving $M > 0$ everywhere; sharp kink at T_c smeared into a crossover; above T_c a Curie–Weiss tail $M = CH_z/(T - T_c) \rightarrow 0$ only as $T \rightarrow \infty$.